

$$X \sim \text{Bernoulli}(p) = \begin{cases} 1 & \text{w.p. } p \leftarrow \\ 0 & \text{w.p. } 1-p \leftarrow \end{cases}$$

Sample $\underbrace{X_1 \dots X_n}_{\text{Outcome of a random experiment}} \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p).$

Goal: Given $X_1 \dots X_n$, $\overset{\text{find}}{\forall} g(X_1 \dots X_n)$ s.t. $p \approx g(X_1 \dots X_n).$

① Maximum Likelihood $\max_{0 \leq p \leq 1} \underbrace{\mathbb{P}(X_1 \dots X_n | p)}_{\text{likelihood}(p)}$

$$\mathbb{P}(X_1 = x_1 | p) = p^{x_1} (1-p)^{1-x_1}$$

$$\begin{aligned} \mathbb{P}(X_1 = x_1 \dots X_n = x_n | p) &= p^{x_1} (1-p)^{1-x_1} p^{x_2} (1-p)^{1-x_2} \dots \\ &\quad p^{x_n} (1-p)^{1-x_n} \\ &\rightarrow p^{\sum_{i=1}^n x_i} (1-p)^{n - \sum_{i=1}^n x_i} \end{aligned}$$

$$\log \mathbb{P}(X_1 \dots X_n | p) = \left(\sum_{i=1}^n x_i \right) \log p + \left(n - \sum_{i=1}^n x_i \right) \log(1-p)$$

$$\frac{d}{dp} \log \mathbb{P}(X_1 \dots X_n | p) = \frac{\sum_{i=1}^n x_i}{p} + \frac{-(n - \sum_{i=1}^n x_i)}{(1-p)} = 0$$

$$\sum x_i - \cancel{\sum x_i} p = np - \cancel{p \sum x_i}$$

$$\rightarrow \hat{p}_{\text{MLE}} = \frac{1}{n} \underbrace{\sum x_i}_{\# \text{ of } 1\text{'s}}$$

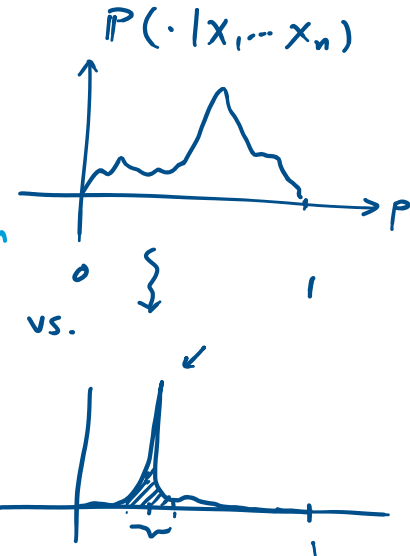
② Bayesian Estimation

P could be seen as a R.V.

$$\Rightarrow P(\overset{A}{p} | \overset{B}{x_1 \dots x_n}) \xrightarrow{\text{likelihood}} \text{prior distribution}$$

$$= \frac{P(x_1 \dots x_n | p) \cdot P(p)}{P(x_1 \dots x_n)}$$

normalizing const. (evidence)



$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

e.g. $x_1 \dots x_n \sim \text{Bernoulli}(p)$

$$P(p) = \text{Beta}(\alpha, \beta) \propto p^{\alpha-1} (1-p)^{\beta-1} \quad \text{if } \alpha, \beta = 1$$

$$P(p | x_1 \dots x_n) = C p^{\sum x_i} (1-p)^{n - \sum x_i} \cdot p^{\alpha-1} (1-p)^{\beta-1} \rightarrow \text{non-info. prior dist.}$$

$$= C p^{\underbrace{\sum x_i + \alpha - 1}_{\alpha'}} (1-p)^{\underbrace{n - \sum x_i + \beta - 1}_{\beta'}}$$

$$= C p^{\alpha'-1} (1-p)^{\beta'-1} = \text{Beta}(\underline{\alpha'}, \underline{\beta'})$$

has a similar structure to the prior density.

Beta ^{is} conjugate for Bernoulli

$$\alpha' = \alpha + \sum x_i$$

$$\beta' = \beta + (n - \sum x_i)$$

How to use Bayesian Estimation in Bandits?



$$R_1 \sim \text{Bernoulli}(p_1)$$

$$\mathbb{E}(R_1) = p_1$$

$$\mathbb{P}(p_1 | D)$$

$$\text{Beta}(1 + \#S_1, 1 + \#F_1)$$

95%

....



$$R_k \sim \text{Bernoulli}(p_k)$$

$$\mathbb{E}(R_k) = p_k$$

$$\mathbb{P}(p_k | D)$$

$$\text{Beta}(1 + \#S_k, 1 + \#F_k)$$

95%

$$i^* = \underset{1 \leq i \leq k}{\text{argmax}} p_i$$

$$\mathbb{P}(p_i) = \text{Beta}(1, 1)$$

$$|D| = n$$

Q: What arm should be selected next?

① Find the high end of the 95% CI of each posterior.

✓ ② Sample the posterior Beta distributions.

$$\hat{p}_1 \hat{p}_2 \dots \hat{p}_k$$

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$$\hat{i}^* = \underset{1 \leq i \leq k}{\text{argmax}} \hat{p}_i$$

→ Tompson Sampling

→ respects the uncertainty that is present in p_i 's.

③ Find the expected val. of the Beta distributions.

$$X \sim \text{Beta}(\alpha, \beta) \quad \mathbb{E}(X) = \frac{\alpha}{\alpha + \beta}$$

$$\mathbb{E}(p_1) = \frac{\alpha + \#S_1}{\alpha + \beta + n_1}, \dots$$

$$\mathbb{E}(p_k) = \frac{\alpha + \#S_k}{\alpha + \beta + n_k}$$